

## ANSWER KEY — Integration Lesson 11 Test: Motion I — From Velocity to Position

For the grader · full working shown · give credit for correct method with small arithmetic slips

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1. A robot rolls at a constant 8 m/s from  $s = 0$ . Where is it after 12 seconds?

**Work:** displacement =  $\int_0^{12} 8 \, dt = 8t \big|_0^{12} = 96$  m. Position =  $0 + 96 = 96$ . **Answer:  $s = 96$  meters.**

2. A car passes mile marker 10 at a steady 50 mph. What marker after 2 hours?

**Work:** displacement =  $50 \times 2 = 100$  miles. Position = starting marker + displacement =  $10 + 100 = 110$ . (Forgetting the starting 10  $\rightarrow$  100 is the classic error.) **Answer: mile marker 110.**

3.  $v(t) = 2t$  m/s; displacement from  $t = 0$  to  $t = 5$ .

**Work:**  $\int_0^5 2t \, dt = t^2 \big|_0^5 = 25 - 0 = 25$ . (Antiderivative check: derivative of  $t^2$  is  $2t$  ✓.) **Answer: 25 meters.**

4. Start  $s = 2$  m,  $v(t) = 3t^2$  m/s; position at  $t = 2$ .

**Work:** displacement =  $\int_0^2 3t^2 \, dt = t^3 \big|_0^2 = 8$ . Position =  $2 + 8 = 10$ . (Check: derivative of  $t^3$  is  $3t^2$  ✓.) **Answer:  $s = 10$  meters.**

5. Start  $s = 20$  m,  $v = -3$  m/s for 4 seconds. Where does it end up?

**Work:** displacement =  $\int_0^4 (-3) \, dt = -3 \times 4 = -12$  m (negative = backward). Position =  $20 + (-12) = 8$ . **Answer:  $s = 8$  meters.**

6.  $v = 5$  m/s on  $[0, 2]$ , then  $v = -2$  m/s on  $[2, 6]$ : (a) displacement, (b) total distance.

**Work:** forward piece:  $5 \times 2 = +10$  m. Backward piece:  $(-2) \times 4 = -8$  m (width  $6 - 2 = 4$  s). (a) Net =  $10 - 8 = +2$  m. (b) Total =  $10 + 8 = 18$  m. **Answer: (a) +2 meters; (b) 18 meters.**

7.  $v(t) = 8 - 2t$ : (a) when does it stop? (b) distance traveled by then.

**Work:** (a)  $v = 0$ :  $8 - 2t = 0 \rightarrow t = 4$  s. (b)  $v$  stays positive before  $t = 4$ , so distance = displacement =  $\int_0^4 (8 - 2t) \, dt = (8t - t^2) \big|_0^4 = 32 - 16 = 16$  m. (Check: derivative of  $8t - t^2$  is  $8 - 2t$  ✓.) **Answer: (a)  $t = 4$  s; (b) 16 meters.**

8.  $v(t) = 4 - 2t$  on  $[0, 4]$ : displacement and total distance.

**Work:**  $v = 0$  at  $t = 2$ . Antiderivative:  $4t - t^2$ . Forward piece  $\int_0^2 = (8 - 4) - 0 = +4$  m. Backward piece  $\int_2^4 = (16 - 16) - (8 - 4) = 0 - 4 = -4$  m. Displacement  $= 4 - 4 = 0$  (it returns to its start!). Total distance  $= 4 + 4 = 8$  m. **Answer: displacement 0; total distance 8 meters.**

9. Graph:  $v = 6$  flat for 3 s, then a line from 6 down to 0 over 2 s. Displacement by geometry.

**Work:** rectangle:  $6 \times 3 = 18$  m. Triangle:  $\frac{1}{2} \times 2 \times 6 = 6$  m. All above the axis, so both count positive. Displacement  $= 18 + 6 = 24$  m. **Answer: 24 meters.**

10. CHALLENGE:  $v(t) = t^2 - 2t$  on  $[0, 3]$ : turnaround, net displacement, total distance.

**Work:** factor:  $v = t(t - 2)$ , zero at  $t = 0$  and  $t = 2$ ;  $v$  is negative on  $(0, 2)$  (test  $t = 1$ :  $1 - 2 = -1$ ) and positive on  $(2, 3)$  (test  $t = 2.5$ :  $6.25 - 5 = +1.25$ )  $\rightarrow$  turnaround at  $t = 2$ . Antiderivative  $F(t) = t^3/3 - t^2$  (check: derivative is  $t^2 - 2t$  ✓).  $F(0) = 0$ ;  $F(2) = 8/3 - 4 = -4/3$ ;  $F(3) = 9 - 9 = 0$ . Piece  $[0, 2]$ :  $-4/3$ . Piece  $[2, 3]$ :  $0 - (-4/3) = +4/3$ . Net  $= -4/3 + 4/3 = 0$  — it ends exactly where it started. Total  $= 4/3 + 4/3 = 8/3$  m. **Answer: turns around at  $t = 2$  s; net displacement 0; total distance  $8/3$  meters ( $\approx 2.67$ ).**